

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

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I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Identification	20%	1.4	1	1	1	1
Debugging	25%	1.6	1	1	1	1.6
Optimization	20%	1	1.4	1.4	1.6	1
Analysis	20%	1	1	1	1	0.4
Code Quality	15%	1	1.6	1.6	1.4	1

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.

- Calculating the total number of usable host addresses after correction.
2. Two devices are configured with the following IPv6 addresses:
- Device A: 2001:db8::ff00:42:8329/64
 - Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.

the IPv4 Network Configuration.

ASWINI-3

IP address: 192.168.10.65

Subnet Mask: 255.255.255.192

Default Gateway: 192.168.10.1

1) Identify the subnet

A/26 subnet has a block size of 64

Subnet ranges:

→ 192.168.10.0 - 192.168.10.63

→ 192.168.10.64 - 192.168.10.127

→ 192.168.10.128 - 192.168.10.191

→ 192.168.10.192 - 192.168.10.255

2) Network Address, Broadcast Address and Host range.

→ Network Address: 192.168.10.64

→ Broadcast Address: 192.168.10.127

→ Valid Host Range: 192.168.10.65 - 192.168.10.126.

3) verify IP address and Gateway.

→ IP Address: valid

→ Gateway 192.168.10.1: Invalid because it belongs
subnet 192.168.10.0/26.

4) Configuration Error.

The workstation and

Subnets. so communication fails.

5) correct configuration:

→ IP Address : 192.168.10.66

→ Subnet Mask : 255.255.255.192

→ Default Gateway : 192.168.10.65

6) Total usable hosts:

$$2^6 - 2 = 62$$

Usable Hosts = 62.

IPv6 Debugging.

Given

Device A: 2001:db8::ff00:42:8329/64

Device B: 2001:db8:0:0:1:1/64

1) Expanded IPv6 Addresses.

Device A:

2001:0db8:0000:0000:0001:0000:0000:0001

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

2) Network prefix:

Device A:

2001:db8:0000:0000::/64

Device B

2001:db8:0000:0000:0001::/64

Network / Prefix:
Device A:

3.) Same Subnet?

No.

Their prefix are different.

4.) correct configuration:

Example:

Device A

2001:db8::ff00:42:8329/64

Device B

2001:db8::10/64

Now both belong to 2001:db8::1/64

5.) Available Hosts:

For /64

$2^{64} = 18,446,744,073,709,551,616$

3) ~~Do~~ Optimize Subnet Allocation using VLSM

Subnet details.

Subnet 1 (/26)

→ Network : 192.168.1.0

→ Broadcast : 192.168.1.63

→ Host Range : 192.168.1.1 - 192.168.1.62

→ Usable Hosts: 62.

Subnet 2 (/27)

→ Network : 192.168.1.64

→ Broadcast : 192.168.1.95

→ Host range : 192.168.1.65 - 192.168.1.94

→ Usable Hosts : 30

Subnet 3 (/28)

→ Network : 192.168.1.96

→ Broadcast : 192.168.1.111

→ Host Range : 192.168.1.97 - 192.168.1.110

→ Usable Hosts : 14

Unused Addresses:

Remaining:

192.168.1.112 - 192.168.1.255

Problems:

→ Some department waste many IP addresses

→ Some departments have insufficient hosts

A IPv6 Debugging:

Given

Device A

2001:db8:1::42:8329/64

Device B

2001:db8:0:0:1::1/64

Addresses:

Device A:

2001:0db8:0000:0000:b000:ff00:0042:8329

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix:

Device A:

2001:db8::/64

Device B

2001:db8:0:0:1:1/64

Verification:

Different prefixes

Hence communication fails.

Correction:

Assign Device B

2001:db8::20/64

Now both device belong to

2001:db8::/64

Routing Table Debugging.

Destination IP

192.168.2.100

(.) Matching Route

Destination belongs to

192.168.2.0/24

2) Next Hop

10.0.0.3

3) Verification Destination

192.168.2.100 lies within

192.168.2.1 - 192.168.2.254

Hence valid.

4) Routing ERRORS.

No routing error exists because the correct route is available

If packets are not reaching the destination verify.

→ Interface status

→ Next-hop availability

→ Routing table configuration

Optimization.

Maintain specific routing entries and ensures the next-hop is active. Use route